

What is energy?

Energy is what makes everything happen. Your body needs energy so that you can move, grow, and keep warm. We also need energy to power our cars, light our homes, and do thousands of other jobs.

Sunshine

We get nearly all our energy from the Sun. Plants absorb the energy in sunlight and store it as chemical energy. The stored energy enters our bodies as food and is released inside our body's cells. All animals and plants obtain their energy from the Sun this way.

Only a tiny fraction of the Sun's energy reaches Earth.

A bow stores energy by bending. When you let go, the bow springs back into shape and releases the stored energy.

Sources of energy

Energy comes from lots of different sources.



Wind drives wind turbines, which convert movement energy into electricity.



Geothermal energy is heat from deep underground.



Plants can be burnt to provide energy for cooking, heating, and lighting.



Waves can be used to make small amounts of electricity.



Dams harness the energy in rivers flowing downhill to make electricity.



The Sun's energy can be captured by solar panels to make electricity.



Fossil fuels such as oil are used to power cars and to make electricity.

Stored energy

An object can store energy and release it later. When you wind up a clockwork toy, energy is stored in a spring. A bow and arrow uses stored energy to shoot the arrow. Stored energy is also called potential energy because it has the potential to make things happen.